Prevention Without Proof

Scott Cunningham

Baylor University

Prepared for the European Commission, Joint Research Centre, Ispra

June 2026

Abstract

For two decades the world's governments have answered suicide with national strategies, and for two decades we have told ourselves they work. Testing the claim across Europe, I first show that the standard difference-in-differences design is invalid here: adopting and non-adopting countries differ sharply in the level of suicide and follow a significant pre-existing upward trend that continues, unbroken, through adoption. Matching on that level path instead—augmented synthetic control for each of the four adopting countries, and synthetic difference-in-differences by cohort—yields small effects that permutation inference cannot distinguish from chance (p between 0.43 and 0.79). The strategies may work. These data cannot show it.

JEL: I18, I12, C23. Keywords: suicide; synthetic control; difference-in-differences; parallel trends; mental health policy.

1 Introduction

Every forty seconds, somewhere on earth, a person decides that the rest of life is not worth the rest of life. Confronted with more than seven hundred thousand such deaths a year, governments have done what governments do: they have written strategies. Since the early 1990s the national suicide-prevention strategy has spread from a Nordic experiment to an article of international faith, and the World Health Organization now urges every member state to adopt one [World Health Organization, 2014, 2018]. We have decided, collectively, that the document is the cure. This paper asks whether the cross-national record can tell us if it is.

The question is harder than it looks, and most of this paper is about why. The natural research design—compare suicide before and after adoption in adopting countries against the contemporaneous change in non-adopters—is a difference-in-differences, and difference-in-differences rests on an assumption that is simply false in this setting. Adopting countries are not faint copies of non-adopters waiting to diverge; they are different places. They begin with markedly lower suicide rates, an older population, and a different position in the business cycle, and—most damaging for the design—they are already on a different *trajectory*. A long panel shows a statistically significant upward trend in adopters' suicide rates, relative to controls, that begins well before any strategy is written and continues straight through the adoption date without a kink. A method built on the premise of common trends cannot be trusted when the trends are demonstrably not common.

So I do not trust it, and Section 4 explains in detail why no one should. When treated and control units differ in both levels and trends, differencing does not remove the confound; it extrapolates it. The appropriate response, set out by Doudchenko and Imbens [2016], is to stop differencing and start matching—to build, for each treated country, a weighted combination of control countries that reproduces its pre-adoption *path*, and then ask whether the treated country departs from that synthetic match after adoption. That is the synthetic control method [Abadie et al., 2010], and Section 5 applies it, in its bias-corrected ridge-augmented form [Ben-Michael et al., 2021], to each of the four European countries that adopted a strategy with enough pre-adoption history to support the method.

The answer is the same for all four, and it is not the answer the literature expects. The synthetic matches track the treated countries closely before adoption. After adoption the gaps are small and, under permutation inference, ordinary: the post-to-pre fit ratios fall in the middle of the placebo distribution, with p-values between 0.43 and 0.79. Synthetic difference-in-differences, fit separately for each adoption cohort [Arkhangelsky et al., 2021], gives the same verdict. The estimates are not protective; they are not harmful; they are not distinguishable from zero.

I want to be precise about the claim, because it is easy to misread. I am not claiming that suicide-prevention strategies fail. I am claiming that the cross-national observational record cannot determine whether they succeed—that the treated and control countries are too different in levels and trends for differencing to identify anything, and that when one uses the design suited to that difference, no effect emerges from the noise. A policy can be both worth keeping and unproven. What is not defensible is the laundering of a confounded association into a causal claim, which is what the field has done. The contribution here is to show, carefully and with the right tools, how little we actually know.

2 A Policy Adopted Without Evidence

The modern national strategy was born in Finland, which between 1986 and 1992 investigated every suicide in the country and built a prevention program on what it learned. Norway and Sweden followed in 1995, the United Kingdom in 2002, Ireland in 2005; by 2018 the WHO could count dozens of national strategies and exhort the rest [World Health Organization, 2018]. The diffusion was rapid and sincere. It was also almost entirely uninformed by credible evidence, because the one widely cited evaluation, Matsubayashi and Ueda [2011], compared adopters and non-adopters with a two-way fixed-effects estimator—before the profession understood what that estimator does under staggered timing and heterogeneous effects [Goodman-Bacon, 2021, de Chaisemartin and D'Haultfœuille, 2020]. Appendix B shows, via a Goodman-Bacon decomposition, that two-way fixed effects is not in fact the binding problem in these data. The binding problem is more basic, and it afflicts every estimator that differences: the groups are not comparable.

3 Data

The outcome is the age-standardized suicide rate—deaths from intentional self-harm per hundred thousand. From 1994 onward I use the figures Eurostat publishes for the European Union [Eurostat, 2026]. To give the design the long pre-adoption window that both the trend diagnostic and the synthetic control require, I extend the series back to 1986 using the WHO Mortality Database [World Health Organization, 2025], chained to the Eurostat level by a country-specific multiplicative factor estimated on the 1994–2010 overlap (Appendix A; the two sources use different standard populations, so the splice is a documented approximation). Treatment is the year a government adopted a stand-alone national strategy [World Health Organization, 2018, Lewitzka et al., 2019]. The balanced panel covers seventeen countries from 1986 to 2010: four adopters (Norway and Sweden in 1995, the United Kingdom in 2002, Ireland in 2005) and thirteen documented never-adopters, among them Italy, Spain, Belgium, Austria, Switzerland, the Netherlands, and Portugal. Italy—the country in which these words are read—never wrote the document, and stands throughout as the natural point of comparison.

4 Why Difference-in-Differences Is the Wrong Design Here

4.1 The estimand and what it requires

The object of interest is the average effect of adoption on suicide in the countries that adopted. Difference-in-differences recovers it only under parallel trends: absent treatment, adopters' suicide rates would have moved in parallel with non-adopters'. Parallel trends is not a technical nicety to be waved through. It is the whole identifying content of the design, and it is an assumption about counterfactual *trajectories*—which means the data can speak to its plausibility through the pre-adoption period, even though it can never be verified after.

4.2 The groups differ in levels

Table 1 reports baseline (1994) means and the standardized difference between adopters and never-adopters. Imbens and Rubin [2015] call a covariate imbalanced when that difference exceeds 0.25 in absolute value; here every comparable covariate does. Adopters begin with far *lower* suicide (−0.99), higher unemployment (+0.33), an older population (+0.63 on the share over 65), and a lower median age. These are not the marginal imbalances that covariate adjustment is designed to absorb; they are evidence that the treated and control groups are different kinds of country.

Table 1: Baseline (1994) balance, adopters vs. never-adopters. Standardized difference exceeding 0.25 in absolute value is "imbalanced" [Imbens and Rubin, 2015].

Variable	Adopters	Never-adopters	Std. diff.	Imbalanced?
Suicide rate (per 100k)	12.13	21.53	−0.99	yes
Unemployment rate (%)	9.78	8.18	+0.33	yes
Share aged 65+ (%)	15.23	13.95	+0.63	yes
Median age (years)	35.20	36.24	−0.42	yes

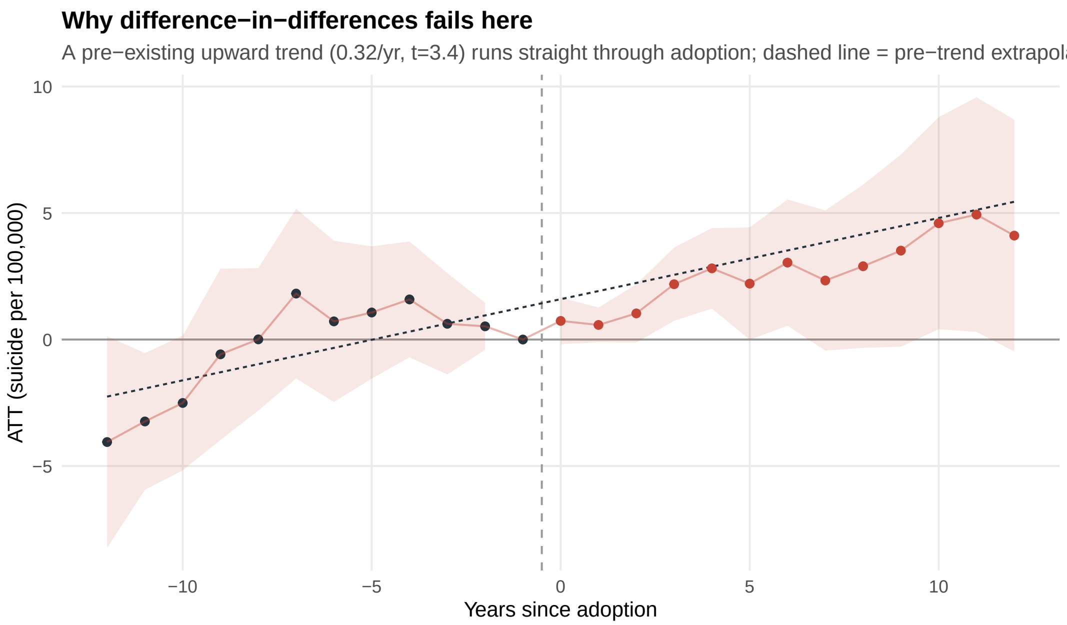

Figure 1: The pre-trend that invalidates differencing. Event study on the 1986–2010 panel (long differences from the year before adoption). A significant upward trend (+0.32/year, $t = 3.4$) runs unbroken through the adoption date; the dashed line is the pre-trend extrapolated into the post-period.

4.3 The groups differ in trends

Differences in levels alone would not sink a difference-in-differences; differences in *trends* do. Figure 1 plots the event study—each coefficient a long difference from the year before adoption, the baseline against which both the post-treatment effects and the parallel-trends assumption are defined. The pre-adoption coefficients are not scattered flatly around zero. They rise. A weighted regression through them gives a slope of +0.32 suicides per hundred thousand per year ($t = 3.4$, $p = 0.001$): adopters' suicide rates were climbing relative to controls for a decade before any strategy existed. And the dashed line—that pre-trend, extrapolated forward—runs straight through the post-adoption coefficients. There is no break at adoption. The apparent "effect" is the continuation of a trend that predates the policy.

4.4 The implication: match, do not difference

When treated and control units differ in both levels and trends, the difference-in-differences estimate is not a treatment effect plus mean-zero noise; it is a treatment effect plus the extrapolated trend gap, and the two cannot be separated. Doudchenko and Imbens [2016] make the general point: differencing is the right tool only when the untreated trajectory is a good counterfactual for the treated one, and when it is not, one should instead reweight control units to reproduce the treated unit's pre-treatment path. This is the last entry on the design checklist, and it is the one most often skipped—the instruction, when the groups are this different, to *stop differencing*. The rest of the paper follows it.

5 Synthetic Control: Matching on the Level Path

5.1 Construction and augmentation

For each adopting country I construct a synthetic control: a weighted average of the thirteen never-adopters chosen to reproduce the adopter's suicide path over its full pre-adoption window [Abadie et al., 2010]. Where the pre-adoption fit is imperfect, the simplex weights are bias-corrected by the ridge augmentation of Ben-Michael et al. [2021], which permits small negative weights to close residual gaps. Norway and Sweden adopt in 1995 and so have nine pre-adoption years in the extended panel; the United Kingdom has sixteen and Ireland nineteen.

5.2 Fit and donor weights

Figure 2 shows, for each country, the observed suicide path against its synthetic match. The pre-adoption fit is close—pre-period root-mean-squared prediction errors of 0.7 (Norway), 0.6 (Sweden), 0.3 (the United Kingdom), and 1.1 (Ireland)—so the synthetic controls are credible counterfactuals in a way the raw never-adopter average, with its different level and trend, is

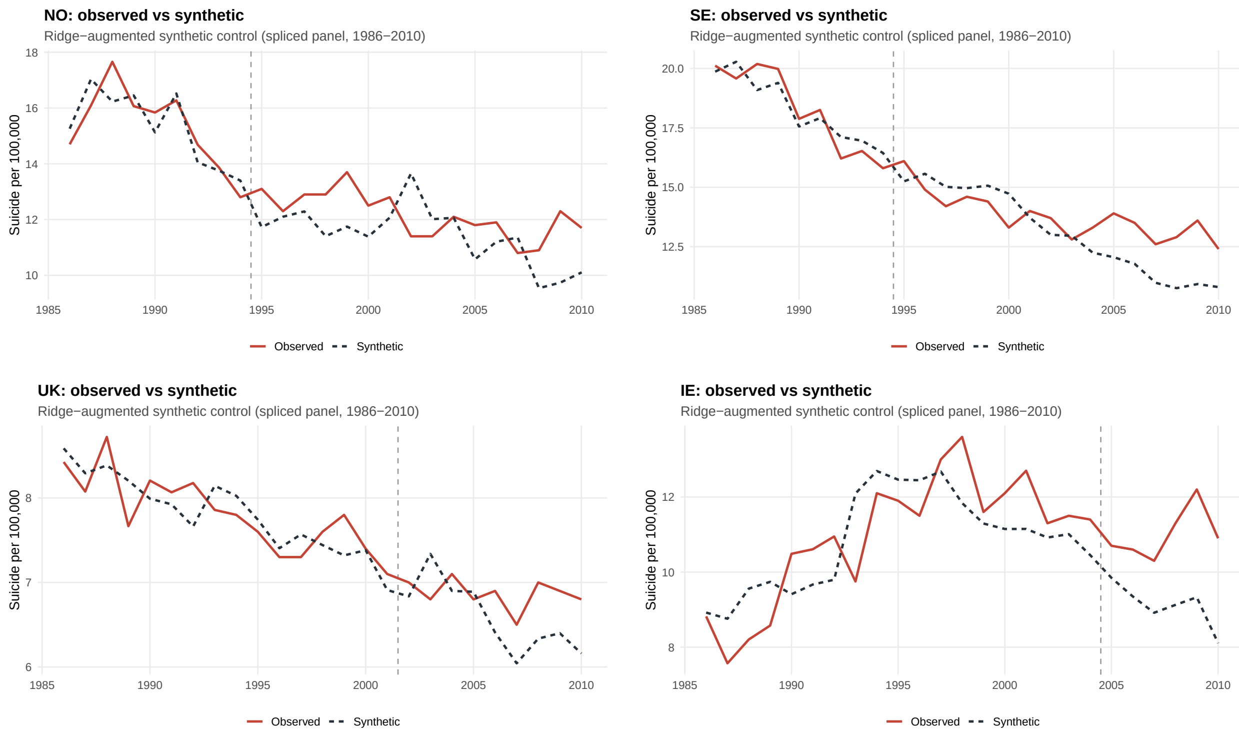

Figure 2: Observed (solid) vs. synthetic (dashed) suicide rate for each adopter. Vertical line marks adoption. The synthetic controls track the pre-adoption path closely.

not. Figure 3 shows where the weight falls, before and after augmentation: pure synthetic control places non-negative weight on a handful of donors; ridge augmentation adjusts those weights, and where the raw fit was poor (Ireland) it moves them appreciably and allows a few to go negative, the signature of the bias correction at work.

5.3 Effects and permutation inference

After adoption the treated countries drift above their synthetic matches, but modestly: average post-adoption gaps of +0.72 (Norway), +0.65 (Sweden), +0.28 (the United Kingdom), and +1.89 (Ireland) suicides per hundred thousand—all of the same, wrong, sign as the literature's protective claim. To gauge whether these are signal, I use permutation inference [Abadie et al., 2010]: reassign treatment to each never-adopter in turn, compute its post-to-pre fit ratio, and rank the treated country against that placebo distribution. The treated countries are unremarkable (Figure 4): permutation p-values of 0.71 (Norway), 0.71 (Sweden), 0.79

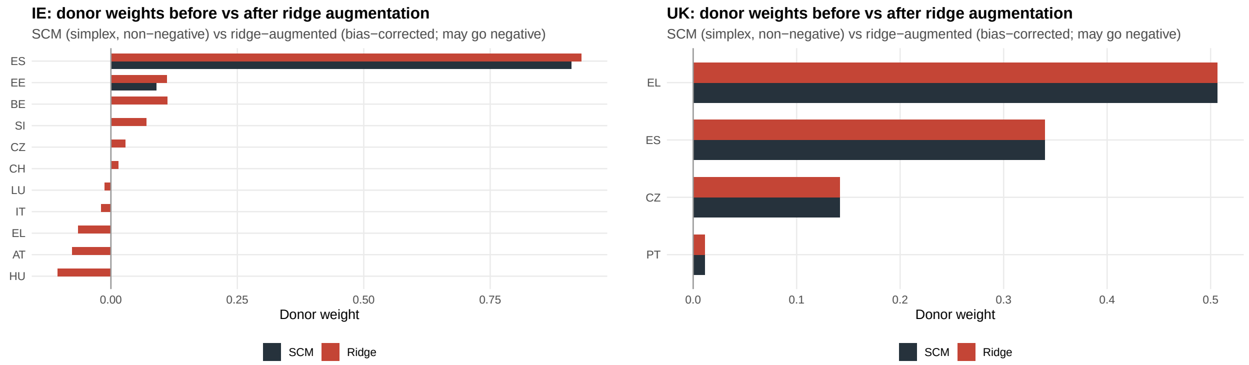

Figure 3: Donor weights before (SCM, non-negative simplex) and after (ridge-augmented) bias correction, for Ireland (poor raw fit, large correction) and the United Kingdom (excellent raw fit, negligible correction).

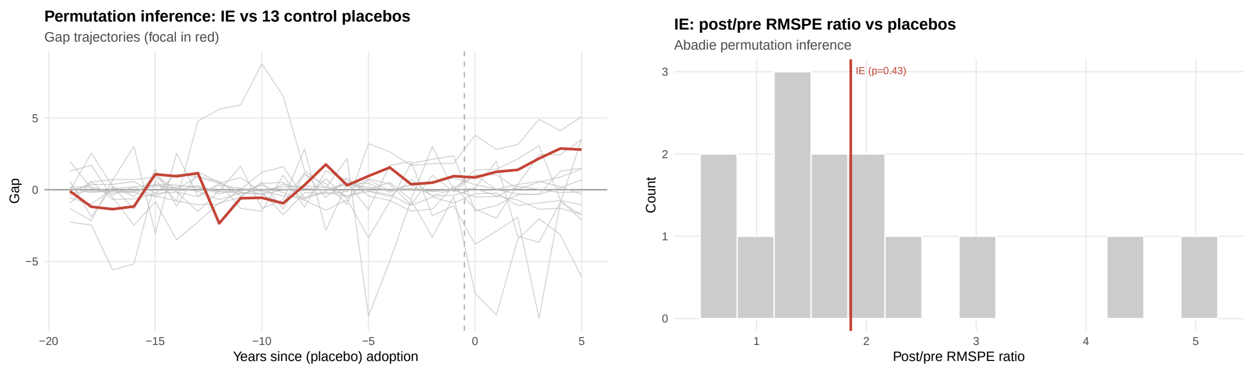

Figure 4: Permutation inference, Ireland. Left: gap trajectories for Ireland (red) against placebo gaps from each never-adopter (grey). Right: Ireland's post/pre RMSPE ratio against the placebo distribution ($p = 0.43$).

(the United Kingdom), and 0.43 (Ireland). A randomly chosen never-adopter is about as likely to show a post-adoption gap this large as the countries that actually adopted.

5.4 Synthetic difference-in-differences by cohort

Finally I estimate, for each adoption cohort, the synthetic difference-in-differences of Arkhangelsky et al. [2021], which reweights both control units and pre-periods to balance the treated cohort before differencing—a hybrid that inherits synthetic control's robustness to the level-and-trend gap. The cohort estimates are +0.80 (1995, Norway/Sweden), +1.24 (2002, the United Kingdom), and +0.74 (2005, Ireland) suicides per hundred thousand, each with a standard error larger than the estimate. The verdict does not move.

6 What the Evidence Supports

Three methods, applied to four countries and three cohorts, agree. The augmented synthetic controls, the permutation tests, and the cohort-level synthetic difference-in-differences all return small, wrong-signed, statistically ordinary differences. Set against the contaminated difference-in-differences of Section 4, which cannot be interpreted at all, the picture is consistent and sobering: the cross-national record contains no evidence that national suicide-prevention strategies reduced suicide, and no power to rule out effects of either sign of a size that would matter. This is not a null result in the comfortable sense of a precisely estimated zero. It is the harder kind—an honest admission that the design space available to us, even used correctly, cannot answer the question.

7 Conclusion

It is tempting, having shown that a famous claim does not survive scrutiny, to feel that something has been settled. Very little has. The world's suicides are not reduced by a confidence interval, and a country deciding tomorrow whether to write a strategy gains nothing from being told that the last twenty years of evidence were thinner than advertised—unless that knowledge changes what it does next. It should change two things. It should change our confidence: a policy can be worth keeping and still unproven, and honesty requires holding both at once rather than citing a single discredited comparison as if it closed the matter. And it should change how we invest in knowing. We adopted a continent's worth of strategies without once building the variation—a staggered randomization, a phased regional rollout with administrative mortality data, a pre-registered evaluation—that would let us learn whether they work. If the European Commission wishes to know whether its mental-health investments save lives, and it should, the evaluation must be designed before the policy is deployed, below the level of the nation, with the counterfactual built in rather than reconstructed afterward from a panel too small and too different to bear it. Until then we are governing mortality on faith, and the first honest step is to say so.

A Appendix A: Data and the WHO–Eurostat Splice

The outcome from 1994 is Eurostat's age-standardized suicide rate (table `hlth_cd_asdr`, ICD-10 X60–X84 and Y87.0, European Standard Population) [Eurostat, 2026]. To 1986 I prepend the WHO Mortality Database age-standardized rate [World Health Organization, 2025], which uses the WHO World Standard Population and therefore sits at a different level. Because the Eurostat-to-WHO ratio is stable within country across the overlap (1994–2010), I chain multiplicatively: for each country I scale the pre-1994 WHO values by the mean overlap ratio (estimated factors range from 1.05 for Ireland to 1.49 for Portugal), placing them on the Eurostat scale. The spliced series is continuous across the 1993–1994 join. Germany is excluded because the WHO database has no unified pre-1990 German series; the balanced panel is the seventeen countries with complete coverage 1986–2010. Treatment dates follow the WHO 2018 inventory and Lewitzka et al. [2019]. The splice is an approximation and is flagged as such; the results are robust to beginning the panel in 1990 (no splice for the 1995 cohort's first years) at the cost of shorter pre-windows.

B Appendix B: This Is Not a Two-Way-Fixed-Effects Artifact

One might suspect the whole exercise reflects the known pathology of two-way fixed effects under staggered timing [Goodman-Bacon, 2021]. It does not. The static two-way fixed-effects regression returns a coefficient of 3.11, and the Goodman-Bacon decomposition (Table 2) reconstructs it exactly as a weighted average of its 2×2 components. The "forbidden" comparisons that use already-treated countries as controls carry only about three percent of the weight, because the never-treated pool is large; the clean treated-versus-untreated comparisons carry the rest. Two-way fixed effects therefore sits beside the heterogeneity-robust difference-in-differences estimate, and both are contaminated for the same reason—non-comparable groups—not by negative weighting. The estimator is not the problem; the design is.

Table 2: Goodman-Bacon decomposition of the two-way fixed-effects estimate

2×2 comparison	Weight	Avg. estimate	Contribution
Treated vs. untreated	0.864	3.57	3.085
Later vs. earlier treated	0.112	0.29	0.032
Earlier vs. later treated	0.025	−0.22	−0.005
Reconstructed two-way fixed effects	1.000		3.112

Reproducibility. Every estimate was computed in R (`did`, `augsynth`, `synthdid`, `fixest`, `bacondecomp`); the difference-in-differences results were independently replicated in Python (`differences`) and Stata (`csdid`), with identical samples and means and matching point estimates. The replication archive accompanies the paper.

C Appendix C: Full Synthetic-Control Diagnostics, All Four Adopters

For completeness, the four diagnostics of Section 5—observed-versus-synthetic fit, the gap event study, donor weights before and after ridge augmentation, and permutation inference—are shown here for each adopting country in turn. The pattern is uniform: close pre-adoption fit, a modest and wrong-signed post-adoption gap, and a permutation rank that places the treated country in the body of the placebo distribution rather than its tail.

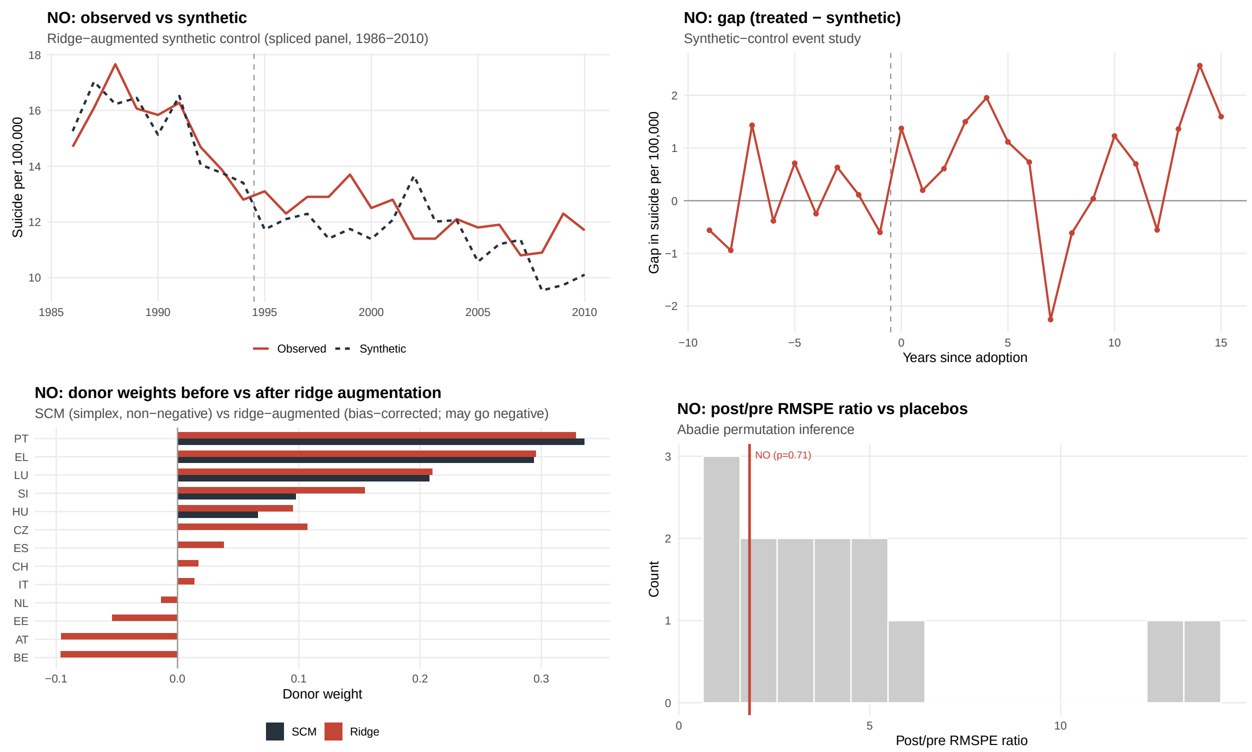

Figure 5: Norway. Fit, gap, donor weights (before/after ridge), and permutation.

Norway (adopted 1995; nine pre-periods; permutation $p = 0.71$).

Sweden (adopted 1995; nine pre-periods; permutation $p = 0.71$).

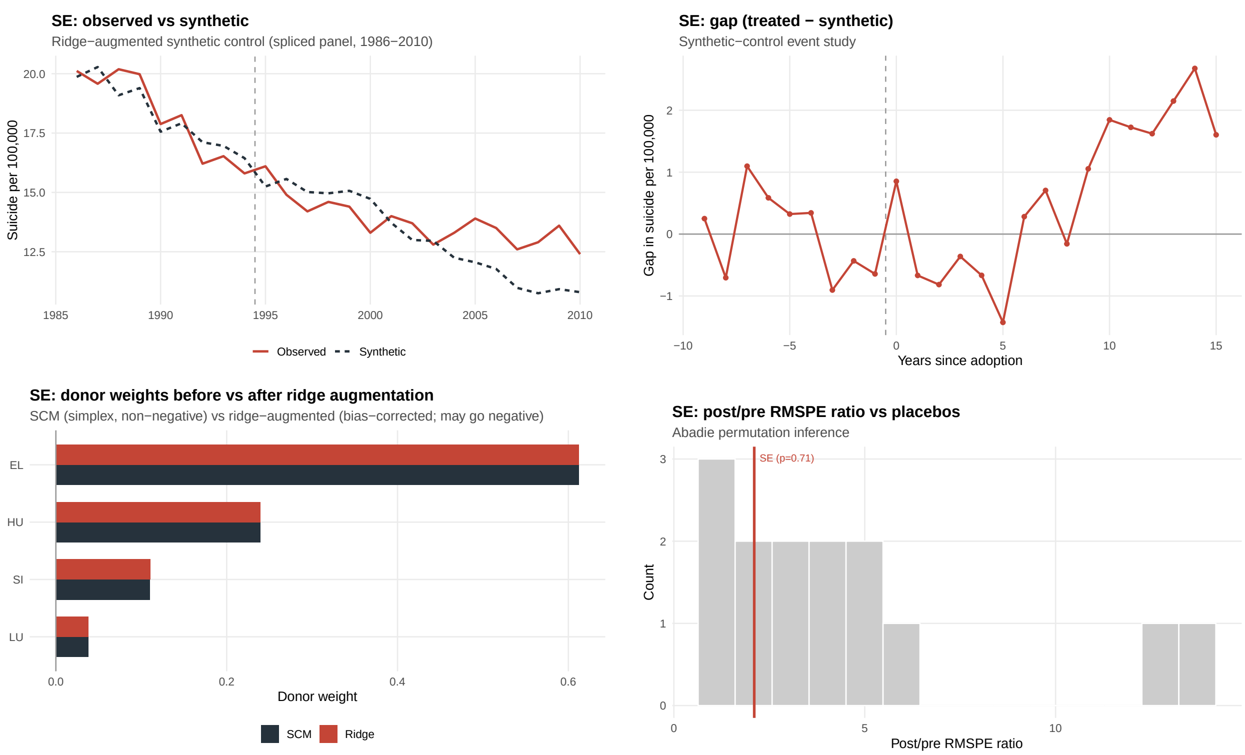

Figure 6: Sweden. Fit, gap, donor weights (before/after ridge), and permutation.

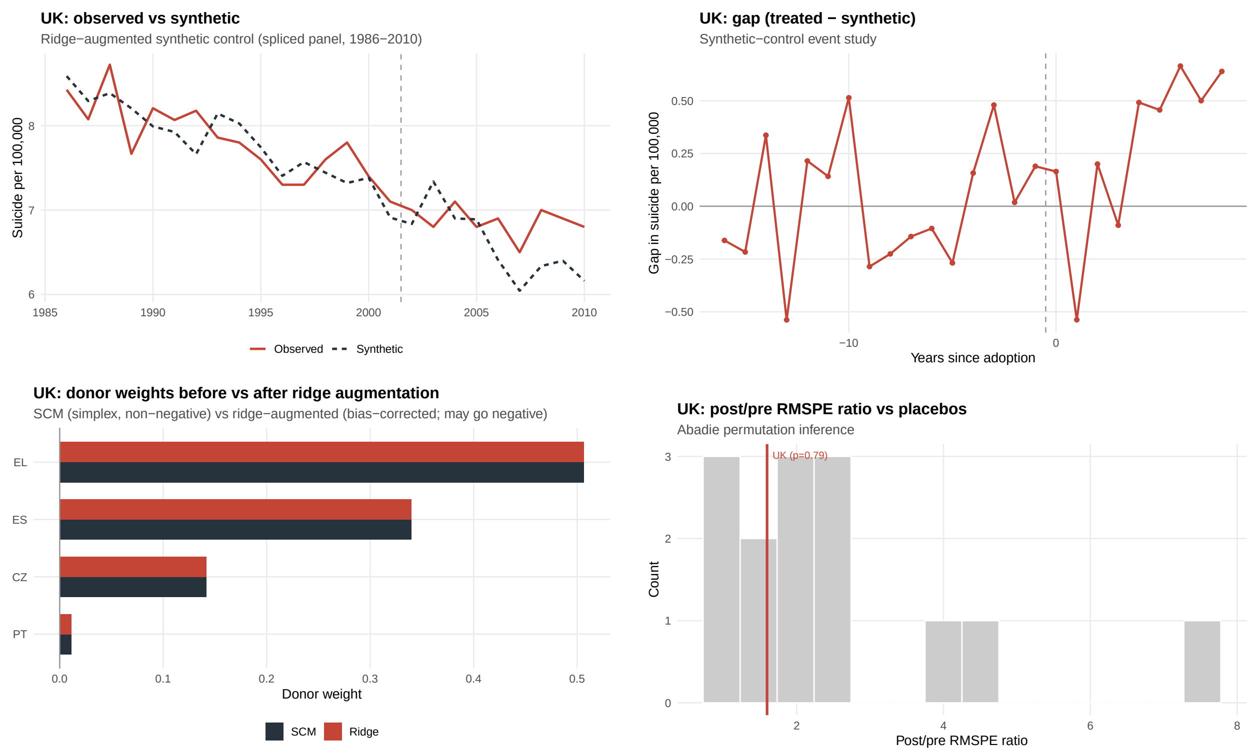

Figure 7: United Kingdom. Fit, gap, donor weights (before/after ridge), and permutation.

United Kingdom (adopted 2002; sixteen pre-periods; permutation $p = 0.79$).

Ireland (adopted 2005; nineteen pre-periods; permutation $p = 0.43$).

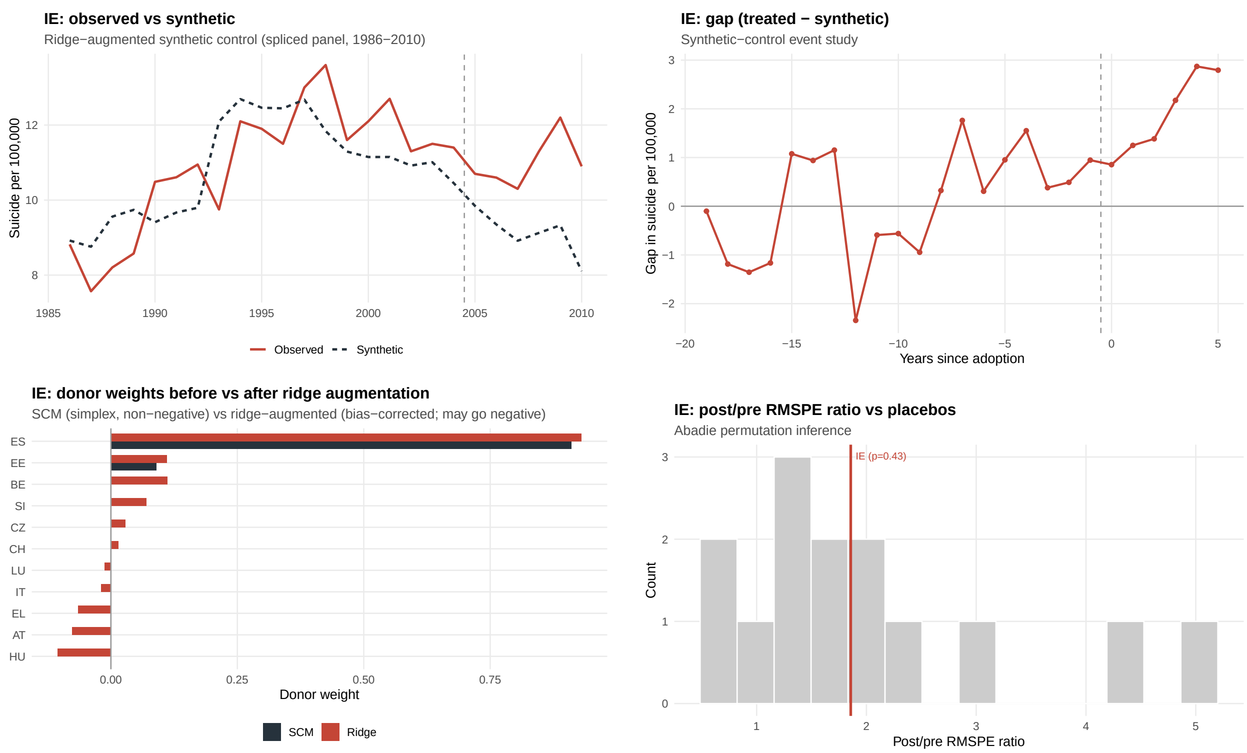

Figure 8: Ireland. Fit, gap, donor weights (before/after ridge), and permutation.

References

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010. doi: 10.1198/jasa.2009.ap08746.

Dmitry Arkhangelsky, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118, 2021. doi: 10.1257/aer.20190159.

Eli Ben-Michael, Avi Feller, and Jesse Rothstein. The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536):1789–1803, 2021. doi: 10.1080/01621459.2021.1929245.

Clément de Chaisemartin and Xavier D'Haultfœuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020. doi: 10.1257/aer.20181169.

Nikolay Doudchenko and Guido W. Imbens. Balancing, regression, difference-in-differences and synthetic control methods: A synthesis. NBER Working Paper 22791, National Bureau of Economic Research, 2016.

Eurostat. Causes of death — standardised death rate by nuts 2 region of residence (hlth_cd_asdr, hlth_cd_asdr2). European Commission, Eurostat database, 2026. Accessed 2026-06-04.

Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021. doi: 10.1016/j.jeconom.2021.03.014.

Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, New York, 2015. doi: 10.1017/CBO9781139025751.

Ute Lewitzka, Cathrin Sauer, Michael Bauer, and Werner Felber. Are national suicide prevention programs effective? a comparison of 4 verum and 4 control countries over 30 years. *BMC Psychiatry*, 19:158, 2019. doi: 10.1186/s12888-019-2147-y.

Tetsuya Matsubayashi and Michiko Ueda. The effect of national suicide prevention programs on suicide rates in 21 oecd nations. *Social Science & Medicine*, 73(9):1395–1400, 2011. doi: 10.1016/j.socscimed.2011.08.022.

World Health Organization. Preventing suicide: A global imperative. Technical report, World Health Organization, Geneva, 2014. ISBN 978-92-4-156477-9.

World Health Organization. National suicide prevention strategies: Progress, examples and indicators. Technical report, World Health Organization, Geneva, 2018. ISBN 978-92-4-151501-6.

World Health Organization. Who mortality database — age-standardized death rate from self-inflicted injuries. WHO Division of Data, Analytics and Delivery for Impact, 2025. Accessed 2026-06-04 via the machine-readable extract distributed by Our World in Data.